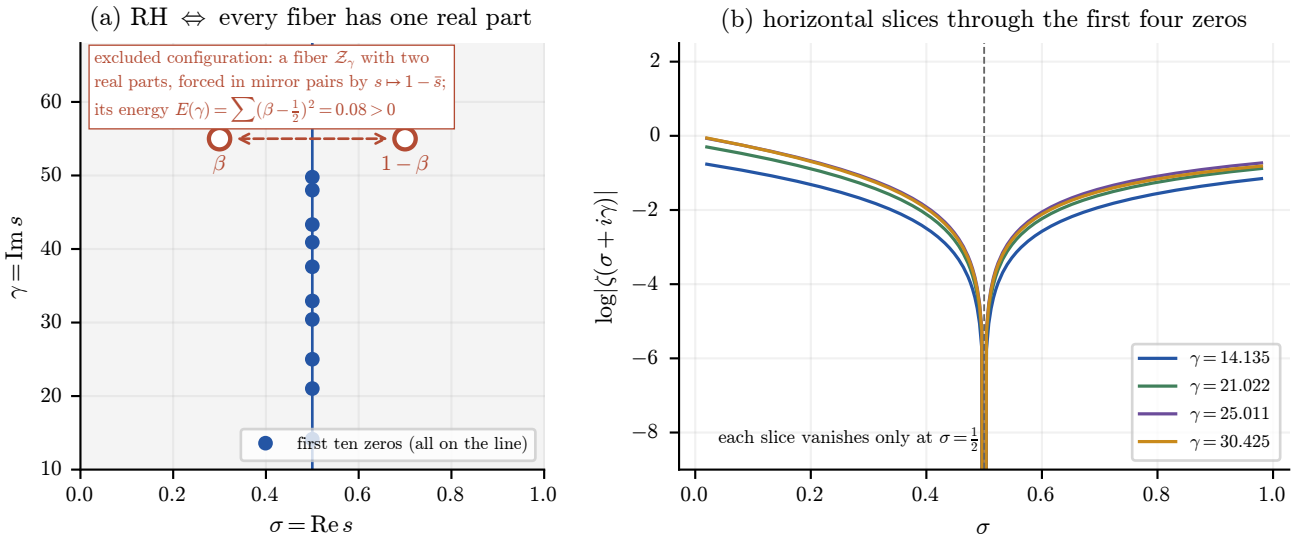
(a) RH ⇔ every fiber has one real part
excluded configuration: a fiber $\mathcal{Z}_\gamma$ with two real parts, forced in mirror pairs by $s \mapsto 1-\bar{s}$; its energy $E(\gamma) = \sum(\beta - \frac{1}{2})^2 = 0.08 > 0$
β
$1-\beta$
first ten zeros (all on the line)
$\gamma = \mathrm{Im}\, s$
$\sigma = \mathrm{Re}\, s$
10
20
30
40
50
60
0.0
0.2
0.4
0.6
0.8
1.0
(b) horizontal slices through the first four zeros
$\log|\zeta(\sigma + i\gamma)|$
each slice vanishes only at $\sigma = \frac{1}{2}$
$\gamma = 14.135$
$\gamma = 21.022$
$\gamma = 25.011$
$\gamma = 30.425$
σ
2
0
−2
−4
−6
−8